

Assignment 1

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Abstract

This report compares three models for sexism detection on English tweets from the EXIST dataset, focusing on Task 2: categorizing sexist messages by author intent into non-sexist (-), direct, judgemental, and reported. We implement a Baseline BiLSTM and a Stacked BiLSTM trained on GloVe embeddings, and fine-tune a Twitter-RoBERTa transformer, evaluating all models over three random seeds using macro F1. The transformer achieves the highest validation F1 (0.634 ± 0.002), outperforming the Stacked BiLSTM (0.387 ± 0.026) and Baseline (0.357 ± 0.025). Error analysis reveals five recurring failure patterns shared across all models, including reported speech confusion, implicit sexism, and inability to handle negation and sarcasm.

1 Introduction

Early text classification approaches relied on SVM or Naive Bayes over bag-of-words features, which cannot capture word order or context. BiLSTMs improved this by processing sequences in both directions, but struggle with long-range dependencies and out-of-vocabulary terms when trained on static embeddings like GloVe. Transformer-based models have since become dominant, offering rich contextual representations and strong transfer learning capabilities.

In this work, we compare three models of increasing complexity on the English subset of the EXIST dataset, evaluated over three random seeds using macro F1 as the primary metric. This progression allows us to assess the benefit of each architectural choice and isolate the contribution of pre-trained representations from that of architectural depth. Several preprocessing functions and training utilities (e.g., `train_model`, evaluation loops) are adapted from course laboratory sessions (NLP and IPCV).

Our main findings: the transformer outperforms both BiLSTM models by a large margin; adding a second BiLSTM layer helps only marginally; all models share common failure patterns that aggregate metrics alone do not reveal.

2 System description

The EXIST dataset contains English and Spanish tweets annotated by six annotators. We apply majority voting to derive a single label per tweet, discard ambiguous cases, and retain only English samples, yielding 2,873 training instances. We observed that label distribution is heavily skewed: the non-sexist class accounts for $\sim 73\%$ of training samples, while DIRECT (20%), REPORTED (7%), and JUDGEMENTAL (5%) are underrepresented. To prevent the models from simply learning to predict the dominant class, we use class-weighted cross-entropy loss, where each class is assigned a weight inversely proportional to its frequency in the training set, so that errors on minority classes are penalized more heavily during training.

For the BiLSTM models, tweets are cleaned via lowercasing, emoji/hashtag/URL removal, and lemmatization. For the transformer, we use the original uncleaned tweets to preserve the informal language the model was pre-trained on.

We implement three models of increasing complexity (Figure 1):

- **Baseline BiLSTM:** The Baseline BiLSTM consists of a GloVe-initialized embedding layer (50-dim), a single Bidirectional LSTM layer (hidden size 128), a dropout layer (0.5), and a fully connected output layer. Out-of-vocabulary tokens (18.71% of the training vocabulary) are assigned small random embeddings. The architecture is adapted from ([CodeGenes.net](https://codegenes.net), 2026).
- **Stacked BiLSTM:** The Stacked BiLSTM extends the Baseline with a second BiLSTM

layer on top of the first, increasing capacity at the cost of a higher overfitting risk.

- **Transformer:** We fine-tune `cardiffnlp/twitter-roberta-base-hate`, a RoBERTa-base model pre-trained on Twitter data for hate speech detection. The original classification head is replaced to map to our four classes, and the HuggingFace Trainer is extended with a custom `compute_loss` method to inject the same class-weighted loss used for the BiLSTM models.

3 Experimental setup and results

All models are trained with seeds 7, 42, 115. The best checkpoint per seed is selected on validation macro F1. BiLSTM models are trained for 100 epochs with Adam ($\text{lr}=1\text{e-}3$), hidden size 128, dropout 0.5. Sequences are padded to a fixed length of 54 tokens, chosen to cover the 99th percentile of training sequence lengths. The transformer is fine-tuned for 5 epochs with AdamW ($\text{lr}=2\text{e-}5$, weight decay 0.01), batch size 8.

Tables 1 and 2 report mean $\pm$ std over seeds on validation and test sets respectively.

Table 1: Validation set results (mean $\pm$ std, 3 seeds).

Model	F1	Prec.	Rec.
Baseline	0.357 $\pm$.025	0.419 $\pm$.047	0.364 $\pm$.018
Stacked	0.387 $\pm$.026	0.467 $\pm$.069	0.379 $\pm$.026
Transformer	0.634$\pm$.002	0.672$\pm$.031	0.625$\pm$.019

Table 2: Test set results (mean $\pm$ std, 3 seeds).

Model	F1	Prec.	Rec.
Baseline	0.351 $\pm$.010	0.385 $\pm$.021	0.367 $\pm$.006
Stacked	0.362 $\pm$.034	0.397 $\pm$.031	0.371 $\pm$.028
Transformer	0.495$\pm$.013	0.517$\pm$.019	0.504$\pm$.010

4 Discussion

Quantitative analysis. The transformer achieves the highest scores across all metrics by a large margin (val F1: 0.634 vs 0.387 and 0.357). Its low standard deviation (± 0.002) confirms that fine-tuning a pre-trained model is far more stable than training from scratch. The Stacked BiLSTM improves only marginally over the Baseline ($+0.030$ val F1), suggesting architectural depth alone provides limited benefit for this task and this is confirmed by precision

and recall curves in Figure 2. All models show a drop from validation to test, most pronounced for the transformer (0.634 $\rightarrow$ 0.495), suggesting some overfitting. The confusion matrices (Figure 3 reveal that all models handle the dominant non-sexist class well, but struggle with minority categories. The transformer stands out on DIRECT (recall 0.69), while JUDGEMENTAL remains the hardest class for everyone.

Error analysis. Manual inspection of misclassified samples reveals five recurring failure patterns:

-*Reported speech confusion:* models cannot distinguish between an author expressing a sexist view and one reporting what someone else said;

-*Implicit sexism:* tweets conveying sexism through implication or personal reflection, without explicit offensive language, are systematically missed;

-*Negation and sarcasm:* when the author’s intent is the opposite of what the words suggest, all models fail;

-*OOV terms:* words not covered by GloVe like *phallocentrism* are treated as noise by the BiLSTM models but handled better by the transformer’s sub-word tokenization;

-*False positives on self-referential content:* tweets where the author uses offensive language to describe themselves are mislabeled as DIRECT by all models, showing that models associate certain words with sexism regardless of who is saying them and in what context.

5 Conclusion

We compared three models for sexism intention classification on English tweets. The transformer clearly outperforms both BiLSTM variants, confirming the advantage of domain-specific pre-training. Between the BiLSTM variants, the Stacked model offers only marginal improvement over the Baseline, suggesting that architectural depth alone is insufficient for this task.

All models share the same core weakness: reliance on surface lexical cues rather than understanding of the author’s intent. Promising directions for future work include data augmentation on minority classes and multi-task learning combining binary sexism detection with intention classification, as the binary label could provide a useful auxiliary signal helping the model first decide whether a tweet is sexist before attempting to classify its intention.

References

CodeGenes.net. 2026. [Understanding and using Py-Torch bi-directional LSTM \(BiLSTM\)](#).

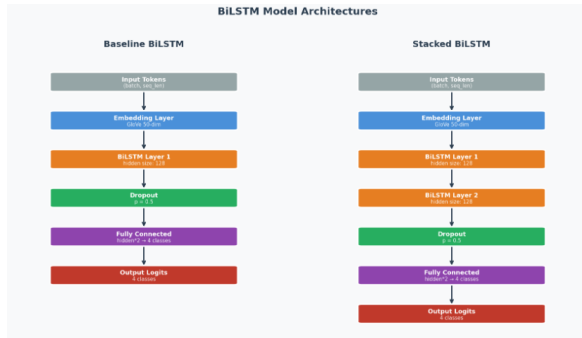


Figure 1: Baseline (left) and Stacked (right) BiLSTM architectures.

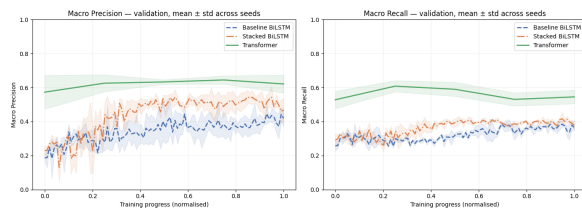


Figure 2: Macro precision and recall on the validation set (mean $\pm$ std across seeds).

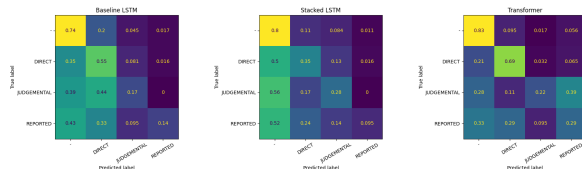


Figure 3: Normalised confusion matrices on the test set